Costs:
- Holding cost: $0.15/croissant/day (applies only to inventory held within shelf life)
- Backlog penalty: $0.30/croissant/day
- No setup/ordering costs
Policy Assumptions:
- Backlogged demand (unmet orders accumulate penalties)
- No storage constraints
- **Friday orders must be placed Thursday** (lead time = 1 day)
Missing Stochastic Detail:
To fully model Fridays, we need:
1. **Demand distribution**.
2. **User's risk tolerance**: Should we optimize for average cost, or guard against extreme spikes (e.g., 99th percentile)?

Actionable Request:
Please provide:
1. The distribution of Friday demand (e.g., \"±20 around 150, uniformly\").
2. Any service-level goals (e.g., \"want to fulfill 95% of Friday demand\").
(This completes the transition from deterministic to stochastic inventory modeling.)

The distribution of Friday demand is ±20 around 150, uniformly,want to fulfill 95% of Friday demand

Normal, variance 30

Base

normal distribution, mean 150, standard deviation 10; guard against extreme spikes

Epoch 1

demand follows a Poisson distribution with mean 150

Epoch 2

Distribution of Friday demand is ±20 around 150, uniformly. Want to fulfill 95% of Friday demand.

Epoch 3

Final Stochastic Inventory Model Summary
1. Demand Structure
- **Weekdays (Mon–Thurs, Sat–Sun)**:
- **Type**: Deterministic
- **Rate**: 120 croissants/day
- **Fridays**:
- **Type**: Stochastic (uniform distribution)
- **Range**: 130–170 croissants (150 ± 20, uniform)
- **Service Level Target**: 95% fulfillment
2. Product & Time Constraints
- **Perishability**: 1-day shelf life (no carryover inventory)
- **Lead Time (L)**: 1 day (Friday orders must be placed Thursday)
- **Time Horizon**: 180 days (~25 Fridays with stochastic demand)
3. Cost Structure
- **Holding Cost**: $0.15/croissant/day (applies only to same-day inventory)
- **Backlog Penalty**: $0.30/croissant/day (unmet demand accumulates daily)
- **Ordering Cost**: $0 (no setup fees)
- **Storage Limits**: None
4. Policy & Optimization Targets
- **Unmet Demand**: Backlogged (penalty applies until fulfilled)
- **Friday Order Quantity**: Must cover the **95th percentile of demand** to meet service level:
- For uniform distribution (130–170), the 95th percentile = **169 croissants** (rounded up from 168.5).
- **Recommended Friday Order**: **169 croissants** (placed Thursday).
5. Baseline Ordering Strategy
- **Non-Fridays**: Order **120 croissants/day** (matches deterministic demand).
- **Fridays**: Order **169 croissants** (covers 95% of stochastic demand).

Key Implications:
- You'll carry **excess inventory on ~5% of Fridays** (when demand < 169), but avoid shortages 95% of the time.
- Perishability prevents holding inventory to offset future shortages.
This fully defines your **perishable inventory system with mixed deterministic/stochastic demand**. Let me know if you'd like adjustments (e.g., tighter cost optimization)!

the model is now clear

I want to know what the probability of a 95% service level is for an order of 150 croissants per day, given a mean of 120 croissants and a standard deviation of 30 croissants

Base

the model is now clear

Epoch 1

the model is clear

Epoch 2

The model is now clear

Epoch 3